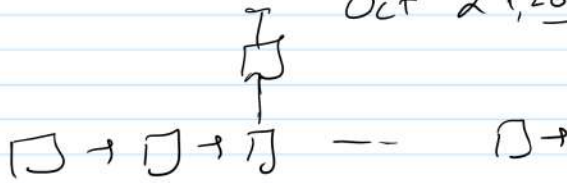


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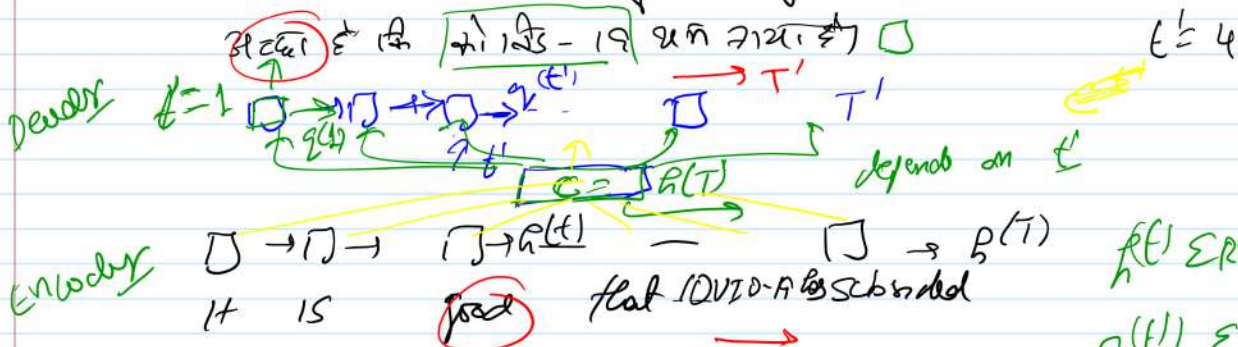
It is good that COVID-19 has subsided
size of the dataset is 19 million words

I:- (Possibility) compute $h(T)$

$$\hat{y} = \underset{\text{softmax}}{\arg\max} (O_0 h(T))$$

II:- seq to seq mapping - $\hat{y}(t') = \underset{\text{softmax}}{\arg\max} (O_0 h^{(t')})$

III:- seq to seq mapping $\hat{y} = (\hat{y}^{(1)} \dots \hat{y}^{(T')})$



$$g^{(t')} = g(O_g g^{(t'-1)} + O_c \cdot c)$$

$$\hat{y}^{(t')} = \underset{\text{softmax}}{\arg\max} (O_0 g^{(t')})$$

$$\text{loss} = \sum_{t'=1}^T L(y^{(t')}, \hat{y}^{(t')})$$

Input:- $(x^{(1)} \dots x^{(T)})$

Output:- $(y^{(1)} \dots y^{(T')})$
 (target)

↳ Idea of Attention:-

$$c^{(t')} = \sum_{t=1}^T \alpha_{t't} h^{(t)}$$

$$\sum_{t=1}^T \alpha_{t't} = 1$$

$h^{(t)} \in \mathbb{R}^d$
 $g^{(t')} \in \mathbb{R}^{d'}$
 $O_g \in \mathbb{R}^{d' \times d}$
 $O_c \in \mathbb{R}^{d' \times d}$

$$a_{t'+1} = U_{ATT}^T \left(U_{ATT} \begin{bmatrix} q^{(t'-1)} \\ p(t) \end{bmatrix} \right) \quad U_{ATT} \in \mathbb{R}^{(1+d) \times d}$$

$$\alpha_{t'} = \text{softmax}(c_{t'})$$

$$\mathcal{L}_{f'} = \{d_{t't}\}_{t=1}^T$$

$$a_4' = \{a_{4t}'\}_{t=1}^T$$